

AYE
TECH HUB

• HVAC ENGINEERING • LESSON 06

Ventilation

Principles

Lesson 06 of 30 | HVAC Engineering Program

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- Why ventilation is more than just fresh air — it removes contaminants
- Natural, mechanical extract, and MVHR systems compared
- ASHRAE 62.1 and EN 13779 ventilation rate requirements
- CO2 monitoring: the most accurate measure of air quality
- Ventilation calculations: air changes per hour and L/s per person

Ventilation — More Than Just Fresh Air

Ventilation is the **controlled movement of air into, through, and out of a building**. Its primary purpose is not comfort — it is health and safety. Ventilation dilutes and removes CO₂ from occupants, moisture from kitchens and bathrooms, volatile organic compounds (VOCs) from furnishings and finishes, and odours throughout the building. Poor ventilation is one of the leading causes of sick building syndrome, mould growth, structural moisture damage, and reduced cognitive performance. A well-designed ventilation system is not a luxury — it is a health infrastructure requirement.

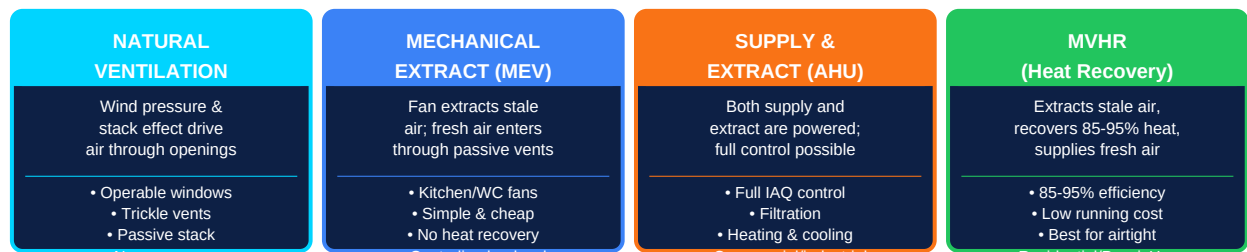


Figure 1 — Ventilation System Types

System Type	Energy Use	IAQ Control	Heat Recovery	Best Application
Natural ventilation	Zero	Low — weather dependent	None	Mild climates; older buildings; low-occupancy spaces
Mechanical extract (MEV)	Very low (extract fan only)	Moderate	None	Kitchens, bathrooms, wet rooms; simple residential
Supply & extract (AHU)	Medium–high (two fans)	High — full control	Optional (run-around/plate)	Offices, commercial buildings, schools, hospitals
MVHR	Low (net) (recover 85–95% heat)	High — continuous	Yes — 85–95% efficiency	Low-energy homes, PassivHaus, airtight new build
Demand-controlled ventilation (DCV)	Low (CO ₂ sensors reduce flow when unoccupied)	Excellent	Combined with MVHR or AHU	Variable-occupancy: offices, classrooms, conference rooms

The airtightness link: Natural and mechanical extract ventilation only work if air can enter the building through gaps. As buildings become more airtight (better for energy), uncontrolled infiltration decreases and deliberate ventilation strategy becomes essential. This is why MVHR is standard in all new PassivHaus buildings.

MVHR — Mechanical Ventilation with Heat Recovery

MVHR is the most energy-efficient ventilation system available for residential and low-energy commercial buildings. It simultaneously extracts stale, moist air from wet rooms (kitchen, bathrooms) and supplies fresh filtered air to living spaces — recovering up to 95% of the heat from the extracted air in the process.

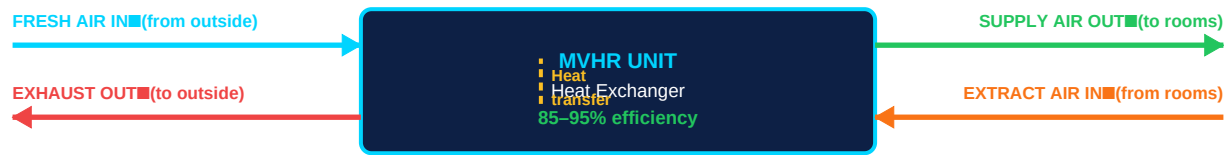


Figure 2 — MVHR Heat Recovery: Fresh Air In, Warm Exhaust Out

MVHR Component	Function	Design Note
Heat exchanger (crossflow/counterflow)	Transfers heat from warm extract air to cold fresh air without mixing the airstreams	Counterflow units achieve up to 95% efficiency vs 75–85% for crossflow
Supply fan	Draws fresh air from outside through a filtered intake, through HX, to supply grilles	Fan must be sized for system resistance — typically 100–200 Pa static pressure
Extract fan	Draws stale air from kitchen, bathrooms, and utility rooms, through HX, to exhaust	Same flow rate as supply fan $\pm$ 10% for slight negative pressure (prevents moisture issues)
Filters (supply side)	F7/ePM1 filter removes pollen, fine particles, and pollution from incoming air	Change every 6–12 months or when pressure drop exceeds specification
Filters (extract side)	G3/coarse filter protects the heat exchanger from cooking greases and lint	Change every 6 months in kitchen extract applications
Summer bypass	In warm weather, bypasses the heat exchanger to provide cool night air without pre-warming	Automatically activated when outdoor temperature < room temperature at night

MVHR duct layout rules: Supply grilles in habitable rooms (bedrooms, living, dining). Extract grilles in wet rooms only (kitchen, bathrooms, utility). Never cross supply and extract ducts. Balance: supply flow = extract flow $\pm$ 10%. Insulate all ducts in cold spaces (loft, unheated spaces) to prevent condensation.

Ventilation Rates, CO₂, and Calculations

CO₂ concentration is the most accurate real-time indicator of ventilation adequacy. Outdoor air is ~420 ppm CO₂. Each person exhales ~200 mL/min of CO₂. As CO₂ rises, cognitive performance declines measurably — studies show a 61% reduction in decision-making ability at 1000 ppm compared to 550 ppm.



Ventilation Rate Requirements

Space Type	ASHRAE 62.1 (L/s per person)	EN 13779 (L/s per person)	Min ACH
Office (open plan)	8.5 L/s/person	8–10 L/s/person	6 ACH min
Classroom	8.5 L/s/person	8–10 L/s/person	6 ACH min
Conference room	8.5 L/s/person	10–12 L/s/person	8 ACH min
Retail / supermarket	8.5 L/s/person + 0.9 L/s/m ²	8 L/s/person	6 ACH
Kitchen (commercial)	N/A (hood-based)	25 L/s/m ²	40+ ACH (hood exhaust)
Hospital patient room	12.5 L/s/person	12 L/s/person	6 ACH minimum
Residential bedroom	6 L/s/person	4–8 L/s/person	0.5 ACH (night)
Bathroom (residential)	N/A	15 L/s extract	15 ACH when occupied

Ventilation Calculations

Calculation	Formula	Example
Air changes per hour (ACH)	$ACH = Q \text{ (m}^3/\text{h)} \div V_{\text{room}} \text{ (m}^3\text{)}$	$Q = 600 \text{ m}^3/\text{h}, V = 300 \text{ m}^3 \rightarrow ACH = 600/300 = 2.0 \text{ ACH}$
Required flow from occupancy	$Q = N_{\text{people}} \times \text{ventilation rate (L/s/person)} \times 3.6$	$20 \text{ people} \times 8.5 \text{ L/s} = 170 \text{ L/s} = 612 \text{ m}^3/\text{h}$
Required flow from floor area	$Q = A_{\text{floor}} \text{ (m}^2\text{)} \times \text{area rate (L/s/m}^2\text{)} \times 3.6$	$200 \text{ m}^2 \text{ office} \times 0.9 \text{ L/s/m}^2 = 180 \text{ L/s} = 648 \text{ m}^3/\text{h}$
Design flow rate	$Q_{\text{design}} = \max(\text{occupancy-based, area-based})$	$\max(612, 648) = 648 \text{ m}^3/\text{h} \rightarrow \text{select AHU for } \geq 650 \text{ m}^3/\text{h}$
CO ₂ check (steady state)	$C_{\text{room}} = C_{\text{outdoor}} + (N \times 0.0052) / Q \text{ (m}^3/\text{s)}$	$20 \text{ people}, Q=0.18 \text{ m}^3/\text{s} \rightarrow C = 420 + (20 \times 0.0052)/0.18 = 998 \text{ ppm (good)}$

What comes next — HVAC-007: Duct Design Basics — how to size supply and return ductwork, the equal friction method, velocity limits for different spaces, duct fittings and pressure loss, and how to read and produce a duct sizing schedule.